

Tutorial

Diode Circuits - Solutions

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Question 1

Let $V(t)$ designate the voltage across the diode and $I(t)$ the current flowing through the circuit. We start by writing two general equations that are always valid:

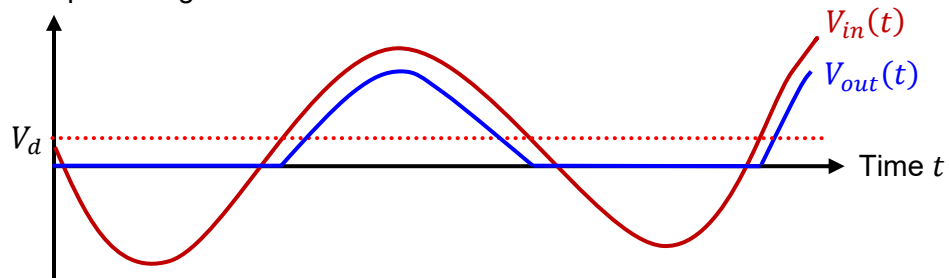
(1) $V(t) = V_{in}(t) - V_{out}(t)$;

(2) $V_{out}(t) = RI(t)$.

We need to take into account the two-state model of the diode by consider two possibilities considered in the table below.

Diode On	Diode Off
With $V(t) = V_d$ and $I(t) > 0$, Equations (1) and (2) become • $V_{out}(t) = V_{in}(t) - V_d$; • $V_{out}(t) > 0$. This inequality implies that $V_{in}(t) - V_d > 0$, i.e., $V_{in}(t) > V_d$.	With $V(t) < V_d$ and $I(t) = 0$, Equations (1) and (2) become • $V_{in}(t) - V_{out}(t) < V_d$; • $V_{out}(t) = 0$. The above inequality implies that $V_{in}(t) < V_d$.

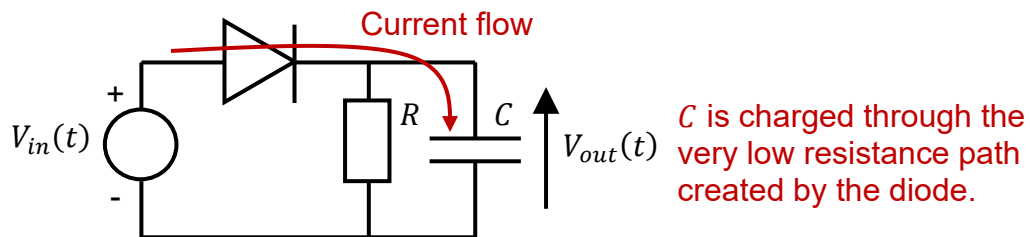
Input and output voltages



Question 2

We start from $V_{in}(t) = 0$. Assuming that the capacitance C is initially discharged, we can also write $V_{out}(t) = 0$.

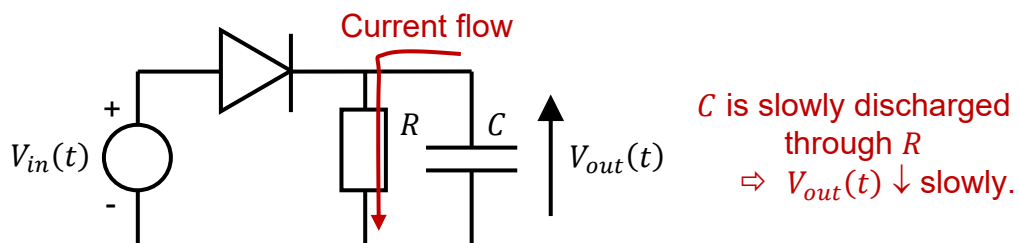
Assume that the input voltage $V_{in}(t)$ increases. As long as $V_{in}(t) < V_d$, the diode remains off and therefore nothing changes in the circuit. Once $V_{in}(t)$ reaches V_d , the diode turns on, and we thus have $V_{out}(t) = V_{in}(t) - V_d$.



The diode turns on when $V_{in}(t)$ becomes greater than or equal to V_d .

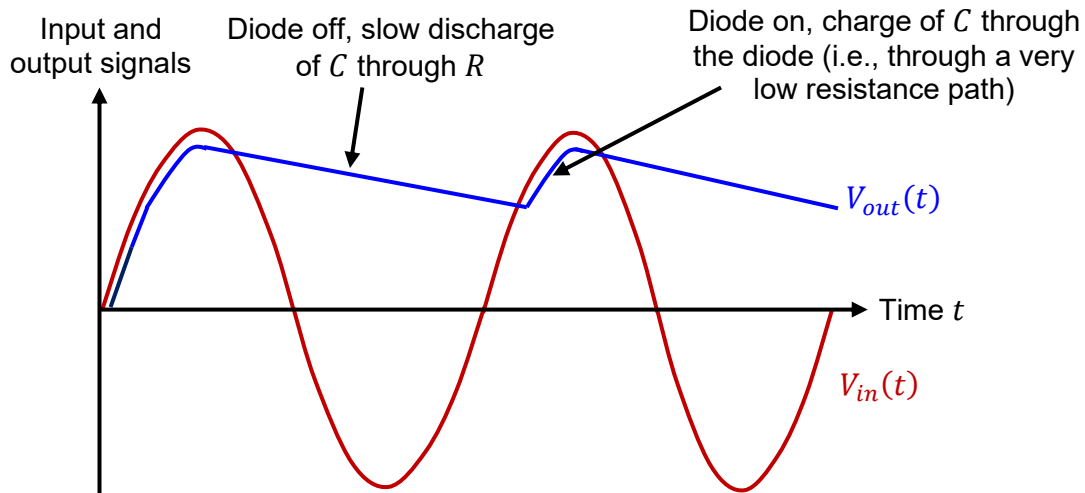
When $V_{in}(t)$ starts falling, $V_{out}(t)$ can no longer follow the change in $V_{in}(t)$ due to the presence of the capacitor-resistor circuit. In fact, the capacitance C cannot be discharged through the diode because the diode current cannot flow in the reverse direction. Instead, the capacitance is discharged through the resistance R because this is the only possible path.

Since this happens slowly (provided that the values of R and C are properly chosen), $V_{out}(t)$ falls more slowly than $V_{in}(t)$. Thus, as soon as $V_{in}(t)$ starts falling, $V_{out}(t) + V_d$ becomes suddenly greater than $V_{in}(t)$, which turns off the diode.



The diode turns off when $V_{in}(t)$ becomes smaller than $V_{out}(t) + V_d$.

At some stage, $V_{out}(t)$ has fallen enough and $V_{in}(t)$ has risen enough again. When that happens, $V_{in}(t)$ becomes once more equal to $V_{out}(t) + V_d$. As a result, the diode turns on again and we then have $V_{out}(t) = V_{in}(t) - V_d$.



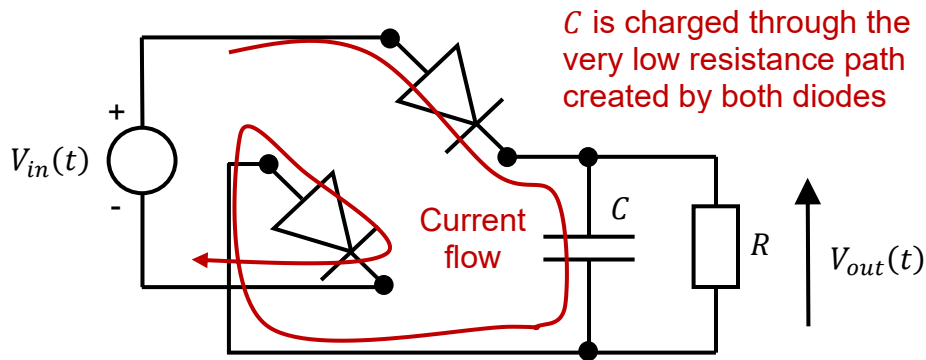
Typical applications of this circuit: power supply (conversion from AC to DC), amplitude demodulation (amplitude detector).

Question 3

We start from $V_{in}(t) = 0$. Assume that the capacitance C is initially discharged and we therefore have $V_{out}(t) = 0$ at the start.

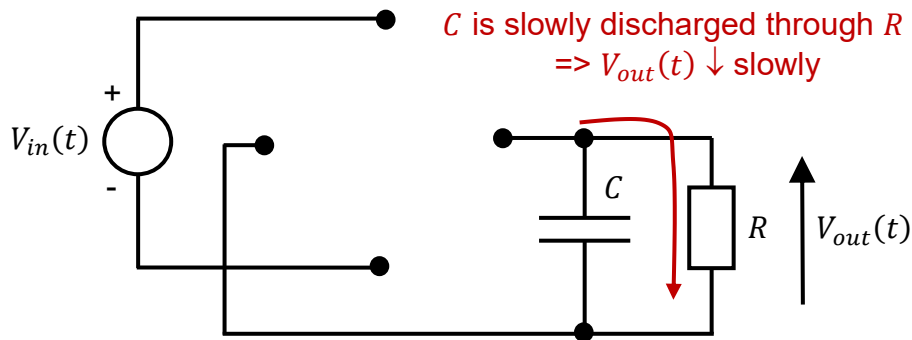
Assume that the input voltage $V_{in}(t)$ increases. As long as $V_{in}(t) < 2V_d$, all diodes remain off and therefore nothing changes in the circuit. Once $V_{in}(t)$ reaches $2V_d$, two diodes turn on, as shown below, and we thus have $V_{out}(t) = V_{in}(t) - 2V_d$.

When $V_{in}(t)$ starts falling, the output voltage $V_{out}(t)$ cannot follow the change in $V_{in}(t)$ due to the presence of the capacitor-resistor circuit. In fact, the capacitance C cannot be discharged through the diodes. Instead, it has to be discharged through the resistance R .



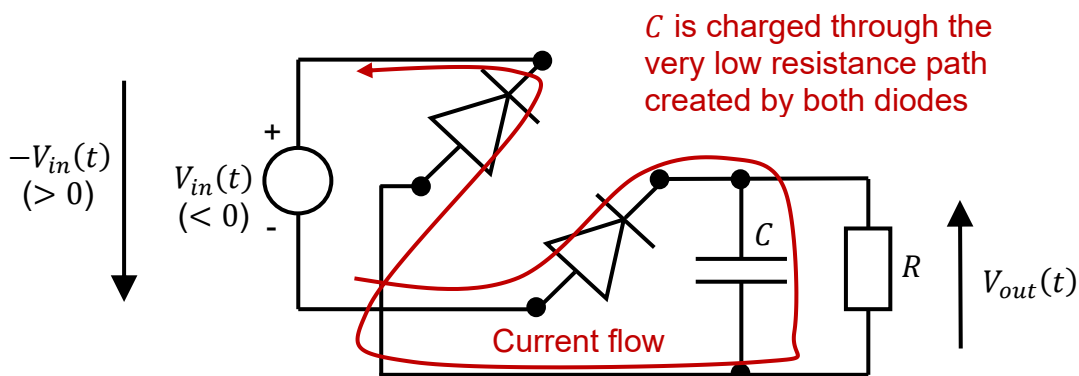
Both diodes turn on when $V_{in}(t)$ becomes greater than or equal to $V_{out}(t) + 2V_d$.

Since this happens slowly, $V_{out}(t)$ falls more slowly than $V_{in}(t)$. Thus, as soon as $V_{out}(t)$ starts falling, $V_{out}(t) + 2V_d$ becomes greater than $V_{in}(t)$, which turns off both diodes.



Both diodes turn off when $V_{in}(t)$ becomes smaller than $V_{out}(t) + 2V_d$.

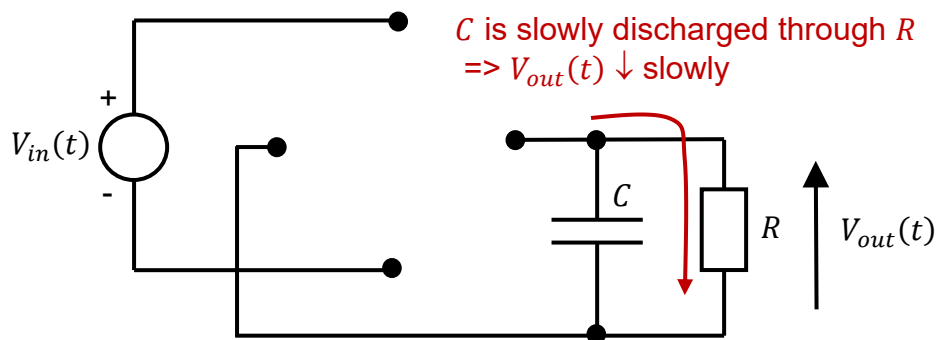
At some stage, $V_{out}(t)$ has fallen enough and $V_{in}(t)$ has also become sufficiently negative. That allows two diodes to turn on, as shown below, and we have $V_{out}(t) = -V_{in}(t) - 2V_d$.



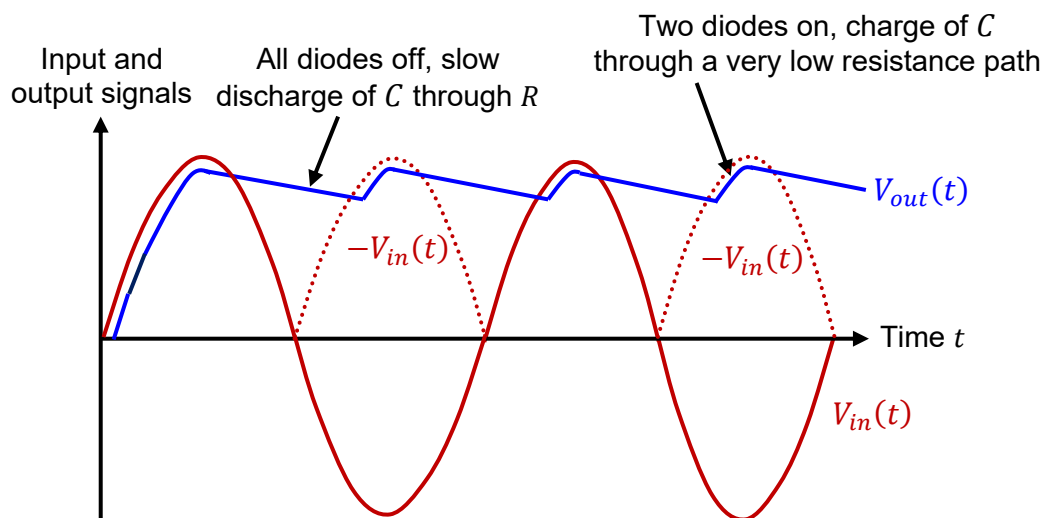
Both diodes turn on when $V_{in}(t)$ becomes sufficiently negative, i.e., when $-V_{in}(t)$ becomes equal to $V_{out}(t) + 2V_d$. When that happens, KVL tells us that $-V_{in}(t) = V_{out}(t) + 2V_d$, i.e., $V_{out}(t) = -V_{in}(t) - 2V_d$.

When $V_{in}(t)$ starts rising again, i.e., becoming less negative, the output voltage $V_{out}(t)$ cannot follow the change in $V_{in}(t)$ due to the presence of the capacitor-resistor circuit.

Once again, the capacitance C is discharged through the resistance R . Since this happens slowly, $V_{out}(t)$ falls more slowly than $-V_{in}(t)$ does. Thus, as soon as $V_{in}(t)$ starts rising, $V_{out}(t) + 2V_d$ becomes greater than $-V_{in}(t)$, which turns off both diodes.



Both diodes turn off as soon as $V_{in}(t)$ starts rising from its most negative value, i.e., $-V_{in}(t)$ becomes smaller than $V_{out}(t) + 2V_d$.



Question 4

Let v_{D1} and v_{D2} designate the voltages across diodes D1 and D2, respectively. Also, let i_1 and i_2 be the currents flowing through diodes D1 and D2, respectively. Finally, let i_{out} denote the current flowing through the resistance R .

We start by writing five general equations:

- (1) $v_{out} = v_{in} - v_{D1}$;
- (2) $v_{out} = v_{in} + v_{D2}$;
- (3) $v_{out} = Ri_{out}$;
- (4) $i_{in} = i_1 - i_2$;
- (5) $i_{out} = (\beta + 1)(i_1 - i_2) = (\beta + 1)i_{in}$.

Combining (4) and (5) leads to $i_{out} = (\beta + 1)(i_1 - i_2) = (\beta + 1)i_{in}$.

Without even considering the two diodes, we have already found the link between input and output currents: $i_{out} = (\beta + 1)i_{in}$.

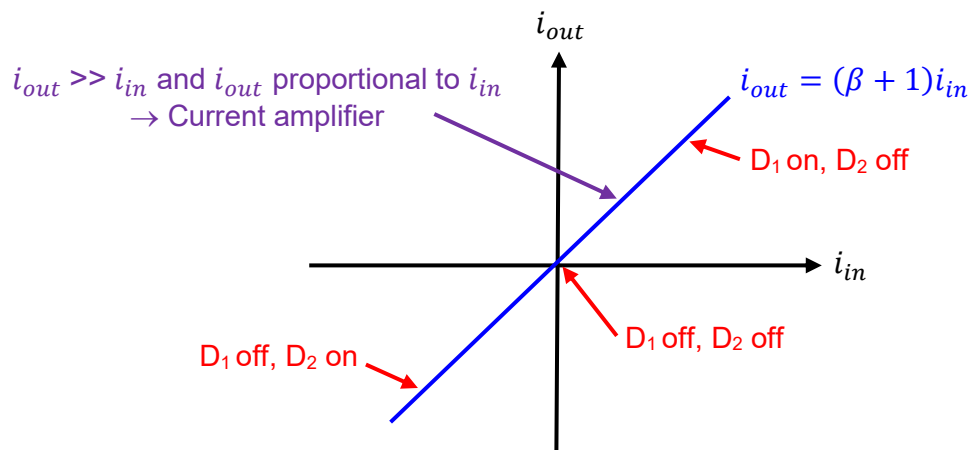
The circuit is a current amplifier with a current gain $A_i = \frac{i_{out}}{i_{in}} = \beta + 1 = 101$.

However, we are now going to consider the diodes in order to find the link between the input and output voltages.

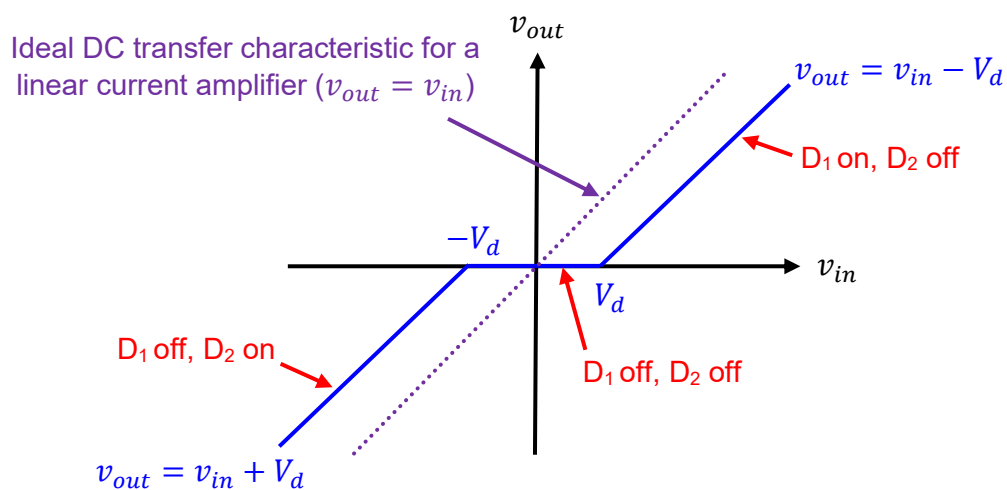
D1 and D2 are off	D1 is on and D2 is off
With $v_{D1} < V_d$, $v_{D2} < V_d$, and $i_1 = i_2 = 0$, the equations yield • $v_{out} > v_{in} - V_d$ • $v_{out} < v_{in} + V_d$ • $i_{in} = 0$, $i_{out} = 0$, and $v_{out} = 0$ $\Rightarrow v_{in} < V_d$ and $v_{in} > -V_d$	With $v_{D1} = V_d$, $v_{D2} < V_d$, $i_1 > 0$, and $i_2 = 0$, the equations yield • $v_{out} = v_{in} - V_d$ • $v_{out} < v_{in} + V_d$ • $i_{in} = i_1 > 0$, $i_{out} = (\beta + 1)i_1 > 0$, and $v_{out} > 0$ $\Rightarrow v_{in} > -V_d$ and $v_{in} > V_d$

D1 is off and D2 is on	D1 and D2 are on
With $v_{D1} < V_d$, $v_{D2} = V_d$, $i_1 = 0$, and $i_2 > 0$, the equations yield • $v_{out} > v_{in} - V_d$ • $v_{out} = v_{in} + V_d$ • $i_{in} = -i_2 < 0$, $i_{out} = -(\beta + 1)i_2 < 0$, and $v_{out} < 0$ $\Rightarrow v_{in} < -V_d$ and $v_{in} < V_d$	With $v_{D1} = V_d$, $v_{D2} = V_d$, $i_1 > 0$, and $i_2 > 0$, the equations yield • $v_{out} = v_{in} - V_d$ • $v_{out} = v_{in} + V_d$ $\Rightarrow$ This is not possible. This case can be discarded.

The DC current transfer characteristic (i_{out} as a function of i_{in}) of the push-pull circuit is shown below.

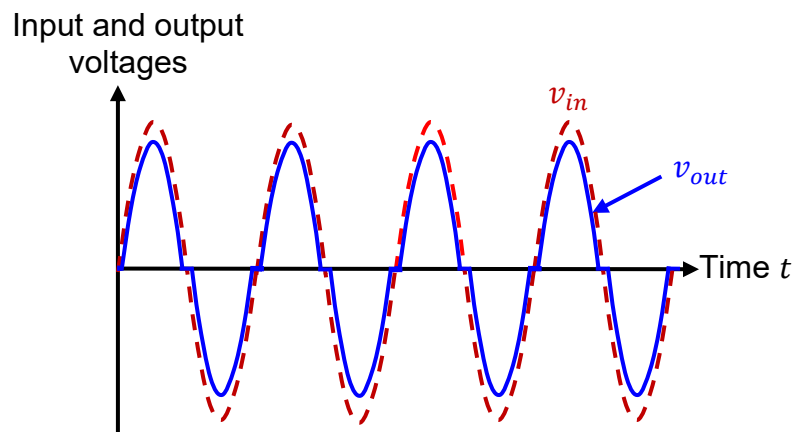


The DC voltage transfer characteristic (v_{out} as a function of v_{in}) of the push-pull circuit is shown below.



We conclude that the push-pull circuit is a current amplifier, but not a voltage amplifier (since v_{out} is not an amplified replica of v_{in}). We have also shown that this circuit is not linear as it introduces some unwanted distortion on the voltage (because $v_{out} \neq v_{in}$). Such distortion, referred to as cross-over distortion, is due to the fact that, for $-V_d < v_{in} < V_d$, both diodes are off, which results in $v_{out} = 0$.

To illustrate the effect of the cross-over distortion, we have plotted below the output voltage v_{out} obtained when the input voltage v_{in} is a sinusoidal signal. For large-amplitude signals, the effect of the cross-over distortion can often be ignored.



Finally, we conclude that this circuit, that can now be referred to as push-pull amplifier, is a non-linear current amplifier.

Question 5

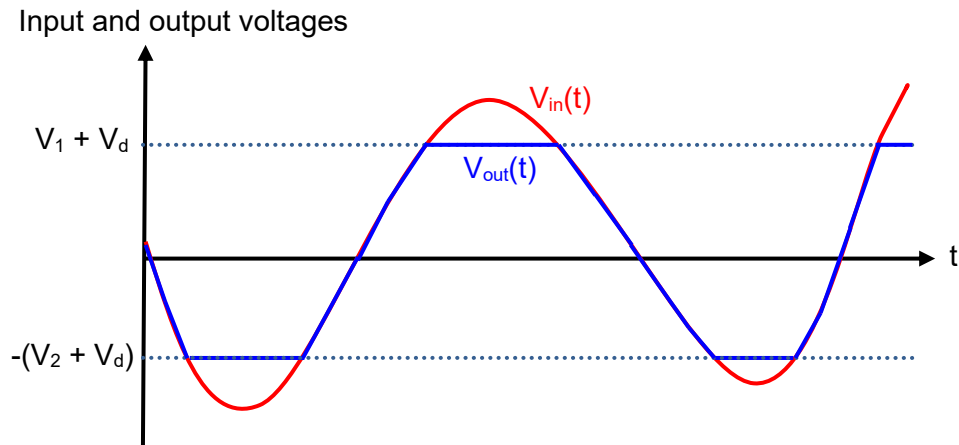
Let $V_{D1}(t)$ and $V_{D2}(t)$ designate the voltages across diodes D1 and D2, respectively. Also, let $i_1(t)$ and $i_2(t)$ be the currents flowing through diodes D1 and D2, respectively. Finally, let $i(t)$ denote the current flowing through the resistor R.

We start by writing four general equations that are always valid:

- $V_{out}(t) = V_1 + V_{D1}(t)$ (eq. 1);
- $V_{out}(t) = -(V_2 + V_{D2}(t))$ (eq. 2);
- $V_{in}(t) = R \cdot i(t) + V_{out}(t)$ (eq. 3);
- $i(t) + i_2(t) = i_1(t)$ (eq. 4).

D1 and D2 are off With $V_{D1}(t) < V_d$, $V_{D2}(t) < V_d$ and $i_1(t) = i_2(t) = 0$, the general equations become • $V_{out}(t) < V_1 + V_d$ • $V_{out}(t) > -(V_2 + V_d)$ • $i(t) = 0 \Rightarrow V_{out}(t) = V_{in}(t)$ $\Rightarrow -(V_2 + V_d) < V_{in}(t) < V_1 + V_d$	D1 is on and D2 is off With $V_{D1}(t) = V_d$, $V_{D2}(t) < V_d$, $i_1(t) > 0$, and $i_2(t) = 0$, the general equations become • $V_{out}(t) = V_1 + V_d$ • $V_{out}(t) > -(V_2 + V_d)$ • $i(t) > 0 \Rightarrow V_{in}(t) > V_{out}(t)$ $\Rightarrow V_{in}(t) > V_1 + V_d$
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D1 is off and D2 is on With $V_{D1}(t) < V_d$, $V_{D2}(t) = V_d$, $i_1(t) = 0$, and $i_2(t) > 0$, the general equations become • $V_{out}(t) < V_1 + V_d$ • $V_{out}(t) = -(V_2 + V_d)$ • $i(t) < 0 \Rightarrow V_{in}(t) < V_{out}(t)$ $\Rightarrow V_{in}(t) < -(V_2 + V_d)$	D1 and D2 are on With $V_{D1}(t) = V_{D2}(t) = V_d$, $i_1(t) > 0$, and $i_2(t) > 0$, the general equations become • $V_{out}(t) = V_1 + V_d$ • $V_{out}(t) = -(V_2 + V_d)$ • This is not possible. This case can thus be discarded.
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We conclude that this circuit performs a clipping function

Question 6

Let V_{D1} and V_{D2} designate the voltages across diodes D1 and D2, respectively. Also, let I_1 and I_2 be the currents flowing through diodes D1 and D2, respectively. Finally, let I denote the current flowing through the resistor R .

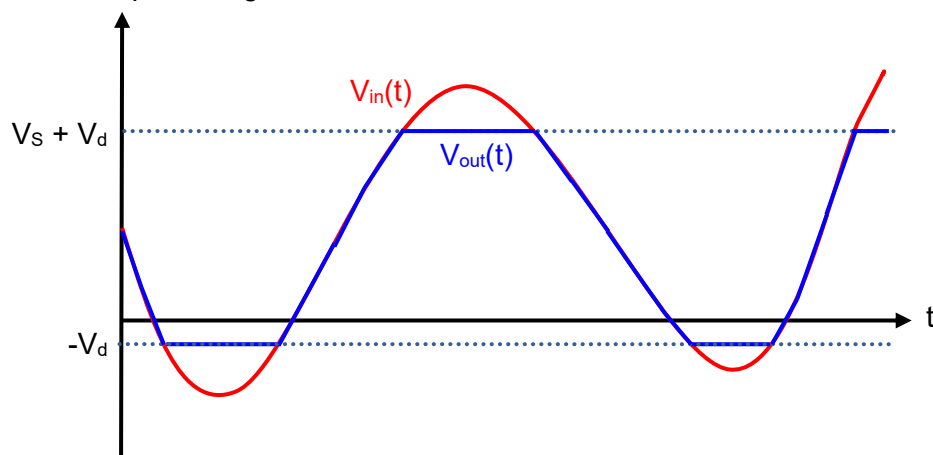
We start by writing four general equations that are always valid:

- $V_{out} = V_{in} - R \cdot I$ (eq. 1);
- $V_{D1} = 0 - V_{out} = -V_{out}$ (eq. 2);
- $V_{D2} = V_{out} - V_S$ (eq. 3);
- $I_2 = I_1 + I$ (eq. 4).

D1 and D2 are off With $V_{D1} < V_d$, $V_{D2} < V_d$ and $I_1 = I_2 = 0$, the general equations become • $I = 0$ • $V_{out} > -V_d$ • $V_{out} < V_S + V_d$ • $V_{out} = V_{in}$ $\Rightarrow -V_d < V_{in} < V_S + V_d$	D1 is on and D2 is off With $V_{D1} = V_d$, $V_{D2} < V_d$, $I_1 > 0$, and $I_2 = 0$, the general equations become • $I = -I_1$ • $V_{out} = -V_d$ • $V_{out} < V_S + V_d$ • $V_{out} = V_{in} + R \cdot I_1 > V_{in}$ $\Rightarrow V_{in} < -V_d$
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D1 is off and D2 is on With $V_{D1} < V_d$, $V_{D2} = V_d$, $I_1 = 0$, and $I_2 > 0$, the general equations become • $I = I_2$ • $V_{out} > -V_d$ • $V_{out} = V_S + V_d$ • $V_{out} = V_{in} - R \cdot I_2 < V_{in}$ $\Rightarrow V_{in} < V_S + V_d$	D1 and D2 are on With $V_{D1} = V_d$, $V_{D2} = V_d$, $I_1 > 0$, and $I_2 > 0$, the general equations become • $V_{out} = -V_d$ • $V_{out} = V_S + V_d$ $\Rightarrow$ This is not possible. This case can be discarded.
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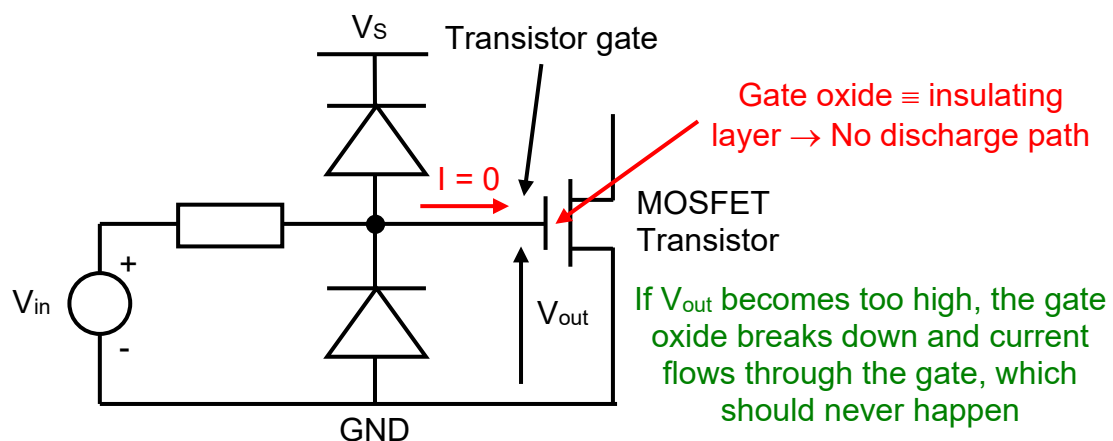
Input and output voltages



The circuit therefore performs a clipping function in the sense that the amplitude of the output signal is always inside the $[-V_d, V_s + V_d]$ range, no matter what the amplitude of the input signal is. Any input signal with amplitudes inside the $[-V_d, V_s + V_d]$ range remains unchanged as it passes through the circuit.

More specifically, this diode circuit is often used to protect another circuit from a harmful phenomenon, known as *electrostatic discharge* (ESD), caused by the discharge of an excess or deficiency of electrons on one surface with respect to another surface or to ground.

ESD takes place when a static charge is transferred from one object to another that is at a different voltage potential. When an ESD-sensitive device, such as a MOSFET transistor, becomes part of the discharge path or is exposed to an electrostatic field, it can be permanently damaged.



For MOSFETs, the gate oxide (insulating layer) between the gate and channel is extremely thin, thus resulting in varying degrees of ESD sensitivity. The electric field E_{ox} across the gate oxide of thickness t_{ox} when a given voltage V is applied is given by $E_{ox} = V / t_{ox}$. For very small t_{ox} , the electric field E_{ox} can exceed the breakdown field of the gate oxide for relatively small voltages. ESD damage of the MOSFET occurs when the gate voltage is high

enough to damage the gate oxide. A leakage path through the gate oxide can then result in a catastrophic failure.

In order to protect a MOSFET from ESD or any other excess gate voltage, the primary objective is to keep the gate voltage from exceeding the breakdown voltage.

Humans can be a source of ESD (e.g., discharge from the fingertip of an individual to a device). A discharge from a device rather than to it can also occur. If a device becomes charged (e.g. sliding down a feeder or plastic tube) and then is discharged by coming in contact or close proximity to a conductive surface, a rapid discharge occurs from the device to the conductive object.

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